

Comparative Study of Non-Invasive Anemia Detection: RGB Analysis of Anterior Conjunctiva versus CIELAB Features from Palm Images using Artificial Neural Networks

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Abstract—This paper presents a comparative study of two non-invasive approaches for detecting iron deficiency anemia. The first approach analyzes RGB color intensity features extracted from anterior conjunctiva images, while the second utilizes CIELAB color space features derived from palm images. The models were implemented using identical ANN architectures with 256–128–64 hidden layers to ensure fair comparison. Experimental results show that the RGB conjunctiva-based method achieved an accuracy of 81%, whereas the palm-based CIELAB approach attained a higher accuracy of 93% with early stopping. The findings indicate that palm image analysis in the CIELAB color space provides superior performance, improved robustness, and greater practical accessibility compared to conjunctiva imaging. Overall, the study highlights the effectiveness of color-based machine learning techniques for non-invasive anemia screening, particularly in resource-limited healthcare settings.

Index Terms—anemia detection, CIELAB, anterior conjunctiva, palm images, artificial neural networks.

I. INTRODUCTION

Iron deficiency anemia is a major global health problem, affecting a significant portion of the population, particularly children and pregnant women in developing regions. According to the World Health Organization, anemia impacts over 40% of children under six years and pregnant women worldwide. Conventional diagnosis relies on invasive blood tests that are costly, time-consuming, and impractical for large-scale screening in resource-limited settings, motivating the need for reliable non-invasive alternatives.

Clinically, pallor of the anterior conjunctiva has long been used as a preliminary indicator of anemia, and recent studies have automated this assessment using image processing techniques. Parallel research has shown that palm images also contain diagnostically relevant color information, offering improved accessibility and ease of acquisition, especially for pediatric populations.

Color-based analysis plays a key role in non-invasive screening. In particular, the CIELAB color space effectively represents human color perception, with the a^* component encoding red–green opponency that correlates strongly with hemoglobin concentration.

In this work, we comparatively evaluate conjunctiva-based RGB analysis and palm-based CIELAB feature extraction

using identical artificial neural network architectures. We aim to objectively assess their diagnostic performance and demonstrate the advantages of palm-based CIELAB analysis for scalable anemia screening.

II. LITERATURE REVIEW

Previous studies have demonstrated the feasibility of non-invasive anemia detection using color analysis of visible anatomical regions. Tamir et al. (2017) proposed an image-based method using anterior conjunctiva photographs, showing that hemoglobin concentration correlates with the red-to-green channel intensity ratio. Their approach achieved 78.9% accuracy, confirming the diagnostic relevance of conjunctival pallor. However, conjunctiva-based imaging faces challenges related to lighting variability, anatomical accessibility, and image acquisition in pediatric populations.

More recent research has explored palm-based imaging as an alternative modality. Appiahene et al. (2023) analyzed palm images converted to the CIELAB color space and reported high classification accuracy using multiple machine learning models. Their study highlighted the strong discriminative capability of the a^* channel, which encodes red–green opponency, showing clear separation between anemic and non-anemic subjects. Palm imaging offers practical advantages, including easier acquisition, standardized region-of-interest extraction, and access to larger public datasets.

Across both modalities, red–green color discrimination consistently emerges as a key indicator of anemia. In RGB-based conjunctival analysis, this is reflected through red-to-green intensity differences, while in the CIELAB color space it is directly captured by the a^* component. These findings provide strong motivation for comparative evaluation of RGB and CIELAB-based approaches under a unified machine learning framework.

III. METHODOLOGY

This work evaluates two non-invasive anemia detection approaches—RGB analysis of anterior conjunctiva images and CIELAB analysis of palm images—under a unified experimental framework. Both methods employ identical feature dimen-

sionality and artificial neural network (ANN) architectures to enable objective comparison.

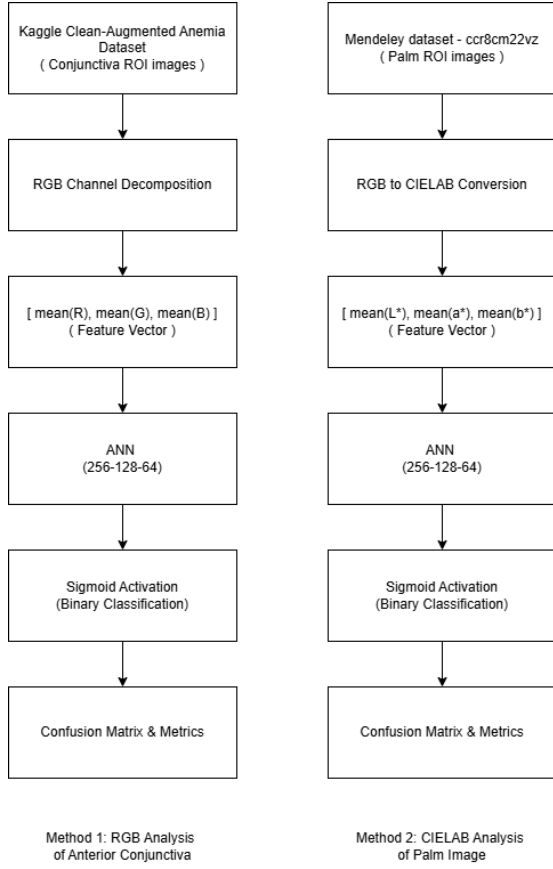


Fig. 1. Comparative methodology framework for non-invasive anemia detection using RGB features from conjunctiva images and CIELAB features from palm images with identical ANN architecture.

A. Method 1: RGB Analysis of Anterior Conjunctiva

1) *Dataset and Image Acquisition:* The Kaggle clean-augmented anemia dataset was used, consisting of smartphone-acquired anterior conjunctiva images with known hemoglobin status. Images were captured under controlled lighting conditions using cameras with a minimum resolution of 12 MP. Regions of interest (ROI) corresponding to the anterior conjunctiva and sclera were manually selected.



Fig. 2. Sample anemic image from the cleaned and augmented Kaggle anemia dataset.

2) *Feature Extraction:* Images were decomposed into individual RGB channels, and the mean pixel intensity of each channel was computed to form a 3-dimensional feature vector:

$$\mathbf{x}_i = [\text{mean}(R_i), \text{mean}(G_i), \text{mean}(B_i)] \quad (1)$$

No additional preprocessing or normalization was applied in order to preserve the direct relationship between color intensity and hemoglobin concentration.

3) *Model Architecture:* A sequential ANN was employed with three input neurons corresponding to the RGB features. The network consisted of three hidden layers with 256, 128, and 64 neurons respectively, using ReLU activation and dropout regularization, followed by a sigmoid-activated output neuron. The model was optimized using the Adam optimizer (learning rate 0.001) with binary cross-entropy loss and trained for 50 epochs with a batch size of 32.

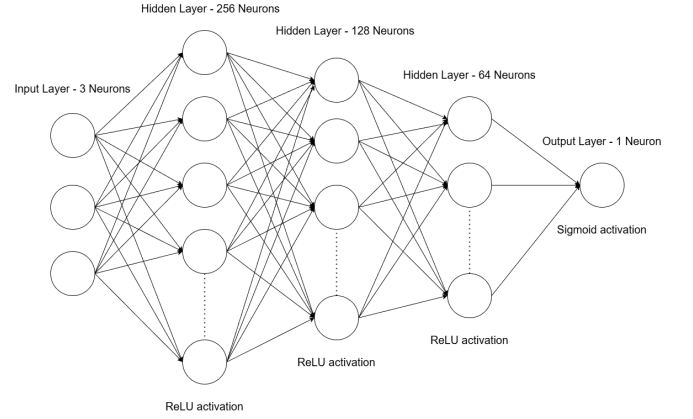


Fig. 3. Artificial neural network architecture.

4) *Evaluation Metrics:* Model performance was evaluated using confusion-matrix-based metrics derived from True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) values:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

5) *Results:* The RGB conjunctiva-based model achieved an overall classification accuracy of 81%. The confusion matrix and detailed performance metrics are presented in Fig. 4 and Table I.

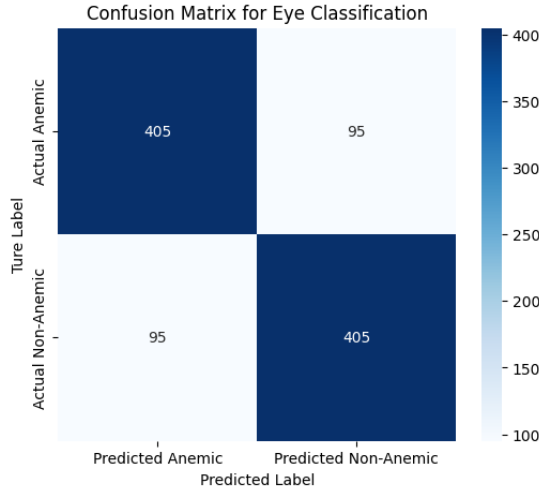


Fig. 4. Conjunctiva Method - Confusion Matrix.

TABLE I
CONJUNCTIVA METHOD - CLASSIFICATION METRICS

Class	Accuracy	Precision	F1-Score	Recall
Anemic (0)	0.81	0.81	0.81	0.81
Non-Anemic (1)	0.81	0.81	0.81	0.81
Weighted Avg	0.81	0.81	0.81	0.81

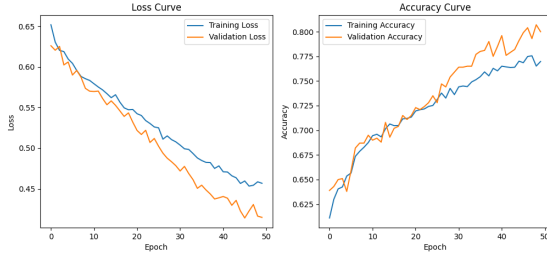


Fig. 5. Training and validation performance curves for the RGB conjunctiva method.

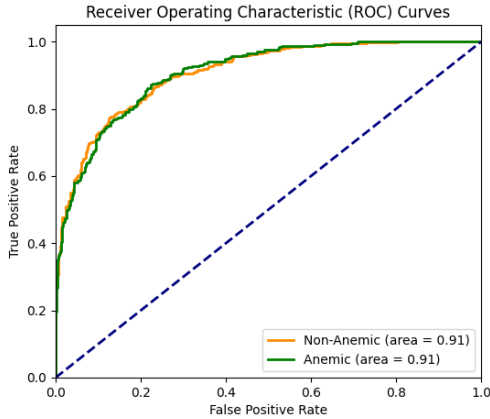


Fig. 6. ROC curve for the RGB conjunctiva method.

B. Method 2: CIELAB Analysis of Palm Images

1) *Dataset and Image Acquisition*: Palm images were obtained from the publicly available Mendeley dataset (ccr8cm22vz), consisting of 527 original images augmented to 2635 samples. Images were captured using smartphone cameras under uniform lighting conditions with flash disabled. Automated thresholding and morphological operations were applied to extract palm ROIs.

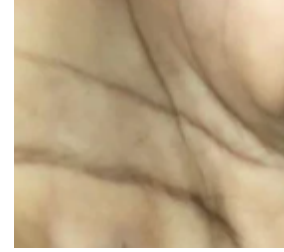


Fig. 7. Sample anemic palm image from the Mendeley dataset.

2) *Feature Extraction*: Extracted palm ROIs were converted from RGB to the CIELAB color space. The mean values of the L^* , a^* , and b^* components were computed to form the feature vector:

$$\mathbf{x}_i = [\text{mean}(L_i^*), \text{mean}(a_i^*), \text{mean}(b_i^*)] \quad (6)$$

Pixel filtering was applied to remove excessively dark or bright pixels, retaining regions suitable for pallor evaluation.

3) *Model Architecture and Training*: The same ANN architecture shown in Fig. 3 was used for fair comparison. Training was performed for up to 150 epochs with a batch size of 16. Early stopping was applied based on validation loss with a patience of eight epochs, using a 70:10:20 train-validation-test split.

4) *Results*: The palm-based CIELAB approach achieved an overall classification accuracy of 93%. The confusion matrix and detailed classification metrics are shown in Fig. 8 and Table II.

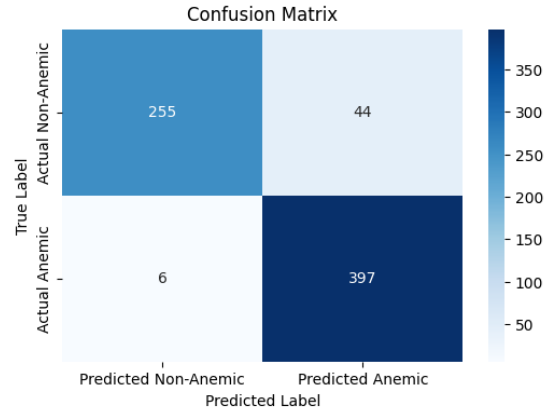


Fig. 8. Palm CIELAB Method - Confusion Matrix.

TABLE II
PALM CIELAB METHOD - CLASSIFICATION METRICS

Class	Accuracy	Precision	F1-Score	Recall
Non-Anemic (0)	0.93	0.98	0.91	0.85
Anemic (1)	0.93	0.90	0.94	0.99
Weighted Avg	0.93	0.93	0.93	0.93

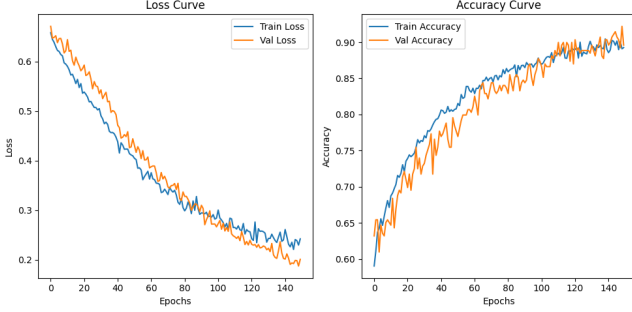


Fig. 9. Training and validation curves for the palm CIELAB method with early stopping.

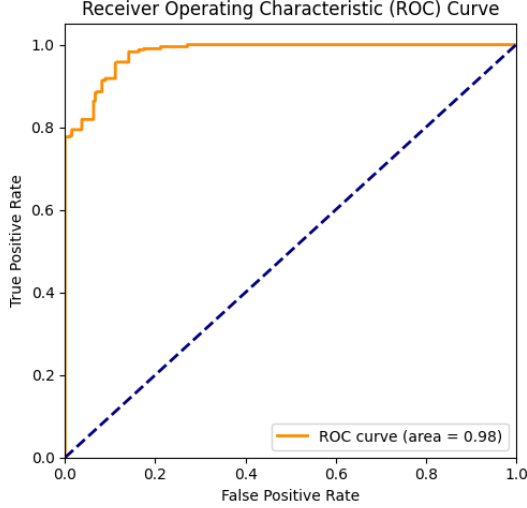


Fig. 10. ROC curve for the palm CIELAB method.

IV. RESULTS AND DISCUSSION

A. Performance Comparison

The palm-based CIELAB method consistently outperformed the RGB conjunctiva approach across all evaluation metrics. As shown in Table III, the CIELAB method achieved an overall accuracy of 93%, representing an 11.56 percentage point improvement over the conjunctiva-based method (81%). Notably, recall improved from 81% to 93%, indicating superior sensitivity in identifying anemic individuals, which is critical for screening applications.

TABLE III
PERFORMANCE COMPARISON: CONJUNCTIVA RGB VS. PALM CIELAB METHODS

Metric	Conjunctiva (RGB)	Palm (CIELAB)
Accuracy	81.00%	93.00%
Precision	81.00%	93.00%
F1-Score	81.00%	93.00%
Recall	81.00%	93.00%

B. Feature Representation and Color Information

Both approaches employ 3-dimensional feature vectors; however, the palm-based method benefits from the perceptually uniform CIELAB color space. In particular, the a^* channel, which encodes red–green opponency, serves as the primary discriminator, analogous to the red–green intensity comparison in RGB analysis. The superior performance of the CIELAB method can be attributed to its robustness against illumination variation, standardized palm image acquisition, and the availability of a larger augmented dataset.

Across both modalities, red–green color discrimination emerged as the dominant diagnostic cue. In conjunctiva images, anemia manifests as reduced red intensity relative to green, while in palm images it appears as lower a^* values. The convergence of these findings across imaging modalities supports the physiological basis of color-based anemia detection.

C. Physiological and Practical Considerations

The effectiveness of both methods is grounded in hemoglobin’s spectral absorption characteristics, which favor green wavelengths over red. Reduced hemoglobin concentration therefore results in diminished red reflectance and a visible shift toward greenish pallor. Both RGB and CIELAB representations capture this phenomenon through different mathematical formulations.

From a practical perspective, palm imaging offers clear advantages in accessibility, ease of acquisition, and patient comfort, particularly for pediatric and elderly populations. Both methods demonstrated low computational overhead, with inference times below 100 ms, enabling real-time screening. Additionally, the availability of a large public palm image dataset enhances scalability and reproducibility compared to conjunctiva-based approaches.

D. Machine Learning Effectiveness and Clinical Implications

The employed ANN architecture (256–128–64) provided sufficient capacity for binary classification while maintaining generalization, aided by dropout regularization and early stopping. The high accuracy achieved by both methods indicates that simple color-based features contain strong discriminatory information for anemia detection, reducing the need for more complex architectures.

Clinically, both approaches offer viable non-invasive alternatives for preliminary anemia screening in resource-limited settings. The higher recall achieved by the palm-based CIELAB method makes it particularly suitable for screening scenarios, where minimizing missed anemia cases is critical.

E. Limitations

Despite promising results, several limitations remain. The datasets originate from different populations and acquisition protocols, complicating direct comparison. Both methods are limited to binary classification and do not estimate anemia severity. Image quality remains dependent on camera hardware and lighting conditions, and external validation on geographically diverse populations is required to confirm generalizability.

V. CONCLUSION

This study demonstrates the effectiveness of non-invasive, image-based anemia detection using color analysis and artificial neural networks. The palm-based CIELAB approach achieved superior performance (93% accuracy and recall) compared to the anterior conjunctiva RGB method (81%), confirming the diagnostic relevance of red-green color information across both imaging modalities.

The advantages of the CIELAB method, including standardized image acquisition, larger publicly available datasets, and improved robustness, make it a strong candidate for scalable anemia screening in resource-limited settings. Nevertheless, conjunctiva-based analysis remains a valuable complementary approach in clinical contexts where ocular assessment is routinely performed.

Future work will focus on external validation across diverse populations, extension from binary classification to continuous severity estimation, multimodal fusion of palm and conjunctiva features, and deployment of smartphone-based screening systems. Overall, the findings support the potential of color-based machine learning techniques for accessible and cost-effective anemia screening in global health applications.

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DATA AVAILABILITY

The datasets used are publicly available:

- Palm images: Asare et al. (2022), Mendeley Data, DOI: 10.17632/ccr8cm22vz.1
- Conjunctiva images: Kaggle Clean-Augmented Anemia Dataset

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